

Notes: Farboodi, Jarosch & Shimer (RES, 2023)

The Emergence of Market Structure

Contents

1	Big picture	2
2	Incremental contribution	2
3	Model primitives and notation	2
4	Equilibrium and theoretical specifications	3
4.1	Lemma 1 panel: Equilibrium trading patterns	5
5	Figures, theorem panels, and equation panels	6
5.1	Figure 1: Direction of trade across contact rates and tastes	6
5.2	Propositions 1 and 2: Rationalizing distributions and continuous heterogeneity	6
5.3	Proposition 3: Linear contact cost and the three equilibrium regimes	7
5.4	Proposition 4: Endogenous middlemen and Pareto tail	7
5.5	Proposition 5: Frictionless limit and persistent intermediation	8
5.6	Planner equations (12)–(15): Social surplus and social value	9
5.7	Figure 2: Equilibrium and optimal counterparty distribution, contact-rate distribution, trading-rate distribution, and misalignment rates	9
5.8	Figure 3: Features of the equilibrium and optimal allocation given relative costs	10
6	Short summary	10
7	Expanded conclusion	11
7.1	Core conclusion	11
7.2	Economic intuition	11
7.3	Incremental contribution	11
7.4	Policy takeaway	11

1 Big picture

- **Question.** Why do over-the-counter markets endogenously develop core-periphery structures and fast intermediaries when traders are ex ante identical?
- **Core mechanism.** Traders choose costly contact rates. Faster traders can intermediate by absorbing misaligned positions from slower traders and offloading them later.
- **Main result.** A rich distribution of contact rates emerges endogenously. Under differentiable costs, the distribution is continuous and has no interior mass points.
- **Middlemen result.** With linear contact costs and an unbounded contact-rate limit, the limiting distribution has a Pareto tail with index 2, and middlemen account for a positive fraction of meetings and trades.
- **Welfare result.** Equilibrium and optimum are qualitatively similar, but equilibrium has excessive contact investment and excessive trading due to congestion and thick-market externalities.

2 Incremental contribution

- **Endogenous heterogeneity from homogeneous agents.** Prior OTC models often assume heterogeneous dealers or contact rates. This paper derives them from ex ante identical traders' investment choices.
- **Intermediation creates heterogeneity.** Earlier work shows heterogeneous access creates intermediation; this paper shows the reverse force too.
- **Analytical Pareto tail.** The paper derives a Pareto tail with index 2 and endogenous middlemen under linear contact costs.
- **Normative benchmark.** It compares decentralized equilibrium with a Pareto optimum and identifies congestion and thick-market externalities.
- **Policy insight.** Optimal interventions reduce excessive contact investment and trading, especially for frequent traders.

3 Model primitives and notation

Traders are ex ante identical, live in continuous time, and choose a contact rate $\lambda \in [0, \bar{\lambda}]$ at birth. Asset holdings are $b \in \{0, 1\}$ and tastes are $i \in \{h, l\}$. Taste shocks arrive at rate γ and exit occurs at rate r . The counterparty distribution overweights fast traders:

$$\mu_F(S) = \frac{\int_S \lambda d\mu_G(\lambda)}{\int_X \lambda d\mu_G(\lambda)}.$$

Interpretation: The probability of meeting a type is proportional to its contact rate, which makes a small number of fast traders important for market outcomes.

The expected value of choosing a contact rate is

$$\pi_\lambda = \frac{1}{2}(v_{\lambda,l,0} + v_{\lambda,h,1}) - C(\lambda).$$

Interpretation: Contact rate is an entry-stage investment into future trading opportunities.

4 Equilibrium and theoretical specifications

Equilibrium is summarized by a counterparty measure μ_F , a surplus function $s(\lambda)$, and a misalignment function m_λ . The key reduced equations are a surplus equation, a flow-balance equation, and an optimal-investment equation.

function (2) as well as the Nash bargaining solution (4) and (5), we derive in [Supplementary Appendix C.1](#) a Bellman equation for the surplus function $s(\lambda)$:

$$(r+2\gamma)s(\lambda) = \Delta + \frac{\lambda}{4} \int_{\mathcal{X}} \left(((s(\lambda') - s(\lambda))^+ - (s(\lambda) + s(\lambda'))^+) m_{\lambda'} \right. \\ \left. + ((-s(\lambda) - s(\lambda'))^+ - (s(\lambda) - s(\lambda'))^+) (1 - m_{\lambda'}) \right) d\mu_F(\lambda'), \quad (8)$$

where $z^+ \equiv \max\{z, 0\}$ and reflects that meetings result in trade if and only if trade is bilaterally efficient. The $r+2\gamma$ on the left-hand side reflects discounting due to exit and taste shocks. Δ is the average difference in flow payoffs between the well-aligned and misaligned states. The remaining terms capture how the option value of trade changes through alignment. At rate $\frac{\lambda}{2}$, traders meet others with opposite asset holdings. When trade occurs, the gains are split equally, giving us $\frac{\lambda}{4}$. The trader meets both misaligned (fraction $m_{\lambda'}$) and well-aligned (fraction $1 - m_{\lambda'}$) counterparties of type λ' . Whenever a trade makes this trader well aligned (misaligned), there is a gain (loss) $s(\lambda)$. Similarly, whenever a trade makes the partner λ' well aligned (misaligned), there is a gain (loss) $s(\lambda')$. Trade occurs only if the sum of the two gains is positive.

We also use the steady state equation (6) and the Nash bargaining solution (4) to derive in [Supplementary Appendix C.1](#) the symmetric flow balance equation governing the stationary distribution of misalignment:

$$\left(r + \gamma + \frac{\lambda}{2} \int_{\mathcal{X}} (\mathbb{I}_{s(\lambda) + s(\lambda') > 0} m_{\lambda'} + \mathbb{I}_{s(\lambda) > s(\lambda')} (1 - m_{\lambda'})) d\mu_F(\lambda') \right) m_{\lambda} \\ = \left(\gamma + \frac{\lambda}{2} \int_{\mathcal{X}} (\mathbb{I}_{s(\lambda) < s(\lambda')} m_{\lambda'} + \mathbb{I}_{s(\lambda) + s(\lambda') < 0} (1 - m_{\lambda'})) d\mu_F(\lambda') \right) (1 - m_{\lambda}). \quad (9)$$

The indicator function $\mathbb{I}$ is equal to 1 if the inequality in the subscript holds and is zero otherwise. The left-hand side captures the outflow from misalignment. Misaligned traders become well aligned if they exit and are replaced, if they have a taste shock, or if they trade. The latter event can occur with both well-aligned and misaligned counterparties, so long as the joint surplus of the transaction is positive. The right hand side captures the inflow into misalignment from well-aligned traders who experience a taste shock or who trade.

Finally, building on equation (3), we prove in [Supplementary Appendix C.1](#) that symmetry and Nash bargaining imply that the choice of contact rate maximizes

$$r\pi_{\lambda} = \delta_1 - \gamma s(\lambda) + \frac{\lambda}{4} \int_{\mathcal{X}} \left((s(\lambda') - s(\lambda))^+ m_{\lambda'} + (-s(\lambda) - s(\lambda'))^+ (1 - m_{\lambda'}) \right) d\mu_F(\lambda') - rC(\lambda), \quad (10)$$

where $\delta_1 \equiv \frac{1}{2}(\delta_{h,1} + \delta_{l,0})$ is the average flow payoff in the well-aligned state. The structure of this equation is similar to equation (8). Equation (10) reflects each trader's cost-benefit analysis when she chooses her contact rate. The first term is the flow payoff from being well aligned. The second term is the cost of losing her well-aligned status following a taste shock. This cost turns out to be decreasing in the trader's contact rate, reflecting the fact that a higher contact rate

Equation panel: reduced equilibrium equations (8)–(10).

Read: Equations (8)–(10) characterize private surplus, stationary misalignment, and contact-rate investment.

Interpretation: These equations compress the dynamic bilateral trading problem into a tractable system over contact rates.

Economic message: Market structure is a fixed point between who traders meet, how misaligned they are, and how profitable it is to invest in contacts.

What not to over-read: These equations are theoretical equilibrium conditions, not empirical regressions.

4.1 Lemma 1 panel: Equilibrium trading patterns

We start by characterizing equilibrium trading patterns given any counterparty measure μ_F .

Lemma 1. *In any equilibrium, the surplus function $s(\lambda)$ is positive-valued and strictly decreasing. When two traders with opposite asset positions meet they*

1. *always trade the asset if both are misaligned;*
2. *never trade the asset if both are well aligned;*
3. *trade the asset if one is misaligned and the other is well aligned and the well-aligned trader has the higher contact rate.*

The Nash bargaining solution (4) and the definition of the surplus function $s(\lambda) = v_{\lambda,h,1} - v_{\lambda,h,0} - q = v_{\lambda,l,0} - v_{\lambda,l,1} + q$ jointly imply

$$p_{\lambda,h,0}^{\lambda',l,1} = \begin{cases} 1 & \text{if } s(\lambda) + s(\lambda') > 0, \\ 0 & \text{otherwise} \end{cases}, \quad p_{\lambda,h,0}^{\lambda',h,1} = \begin{cases} 1 & \text{if } s(\lambda) > s(\lambda'), \\ 0 & \text{otherwise} \end{cases},$$

$$p_{\lambda,l,0}^{\lambda',h,1} = \begin{cases} 1 & \text{if } 0 > s(\lambda) + s(\lambda'), \\ 0 & \text{otherwise} \end{cases}, \quad p_{\lambda,l,0}^{\lambda',l,1} = \begin{cases} 1 & \text{if } s(\lambda') > s(\lambda), \\ 0 & \text{otherwise} \end{cases}.$$

Lemma 1: equilibrium trading patterns.

Read: Misaligned pairs trade, well-aligned pairs do not, and fast traders intermediate for slower traders.

Interpretation: Fast traders become intermediaries because their future trading opportunities make temporary misalignment less costly.

Economic message: Static efficiency creates fundamental trades; dynamic option value creates intermediation trades.

What not to over-read: The lemma conditions on a counterparty distribution; it does not yet determine that distribution.

5 Figures, theorem panels, and equation panels

5.1 Figure 1: Direction of trade across contact rates and tastes

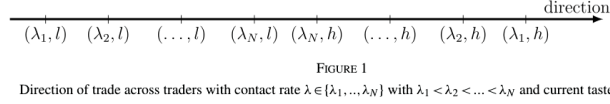


Figure 1: direction of trade across contact rates and tastes.

Read: Fast traders lie near the center of the chain and trade in both directions. Slow traders at the periphery trade mainly when misaligned.

Interpretation: The endogenous core consists of traders whose value comes from quickly unwinding positions.

Economic message: Intermediation is dynamic inventory reallocation toward agents with better future contact opportunities.

What not to over-read: The figure is a reduced representation of trading direction, not a literal network graph.

5.2 Propositions 1 and 2: Rationalizing distributions and continuous heterogeneity

Proposition 1. *For any counterparty measure μ_F , there exists a cost function C such that μ_F is an equilibrium. Moreover, C is unique on support of μ_F , up to an additive constant.*

The formal proof proceeds in three steps. First, we show that the counterparty distribution uniquely determines the misalignment rate. Then, we derive the functional form for the surplus equation, thereby proving that the counterparty distribution uniquely determines the surplus function. Finally, we show how to recover the cost function from these three objects.

Since every contact rate measure μ_G is associated with some counterparty measure μ_F , a corollary is that our model can rationalize any contact rate measure through the choice of an appropriate cost function C . As such, our framework offers a novel way of modelling and rationalizing real-world trading networks, complementing the existing literature on network formation. In particular, we can rationalize the coexistence of traders with very different rates of market access despite them being *ex ante* identical. In the next subsection as well as Section 6, we invert the analysis in Proposition 1 to characterize the counterparty distribution under some natural assumptions on the shape of the cost function.

5.3. Heterogeneity in contact rates

We next show our second main result, that the coexistence of traders with different contact rates arises naturally in equilibrium even when market participants are *ex ante* homogeneous.

Proposition 2. *Assume C is differentiable and C' is Lipschitz continuous. Then any equilibrium counterparty distribution F and contact rate distribution G are absolutely continuous on $[0, \bar{\lambda})$. If additionally C is weakly convex, $C'(0) < \frac{\Delta \gamma^2}{2r(r+2\gamma)^3}$, and $C'(\bar{\lambda}) \geq \frac{4\gamma\Delta}{r\bar{\lambda}^2}$, then a positive measure of traders choose a contact rate in the interval $(0, \bar{\lambda})$ in any equilibrium.*

Propositions 1 and 2.

Read: Proposition 1 rationalizes any counterparty measure using a suitable cost function. Proposition 2 shows that smooth costs imply continuous dispersion.

Interpretation: Intermediation creates local incentives to deviate from mass points, preventing interior atoms in the contact-rate distribution.

Economic message: The model can generate tiered OTC networks without assuming ex ante dealer types.

What not to over-read: Proposition 1 is a rationalization result, not a claim that every distribution is equally plausible under one fixed cost function.

5.3 Proposition 3: Linear contact cost and the three equilibrium regimes

We start by proving existence of equilibrium and characterizing its properties when the cost function is linear:

Proposition 3. Assume $C(\lambda) = c\lambda$. Fix r , γ , Δ , and $\bar{\lambda}$. There exist thresholds $\bar{c} > \underline{c} > 0$ such that

$$\text{if } \begin{cases} c \geq \bar{c} \\ c \in (\underline{c}, \bar{c}) \\ c \leq \underline{c}, \end{cases} \quad \text{then there is a } \begin{cases} \text{autarky equilibrium} \\ \text{intermediated trade equilibrium} \\ \text{degenerate trade equilibrium,} \end{cases}$$

and any equilibrium takes one of these three forms. In an autarky equilibrium, the average contact rate is $\Lambda = 0$. In an intermediated trade equilibrium, the average contact rate is $\Lambda \in (0, \bar{\lambda})$; the support of the counterparty distribution is a convex interval $[\underline{\lambda}, \bar{\lambda}]$ with $\underline{\lambda} \in (0, \bar{\lambda})$ and $dF(\bar{\lambda}) > 0$; and the misalignment rate m_λ is increasing on $[\underline{\lambda}, \bar{\lambda}]$. In a degenerate trade equilibrium, the average contact rate is $\Lambda = \bar{\lambda}$.

Proposition 3.

Read: Depending on costs, equilibrium can be autarkic, intermediated, or degenerate.

Interpretation: The realistic intermediated region appears when contacts are neither too costly nor too cheap.

Economic message: A nontrivial OTC hierarchy emerges in the intermediate-cost region.

What not to over-read: The thresholds are theoretical objects, not estimates for a specific market.

5.4 Proposition 4: Endogenous middlemen and Pareto tail

Proposition 4. Assume $C(\lambda) = c\lambda$ with $c < \frac{\gamma\Delta}{8r(r+\gamma)(r+2\gamma)}$. Then in a limiting equilibrium, there are middlemen, meaning $\lim_{\lambda \rightarrow \infty} F(\lambda) < 1$; and the contact rate distribution has a Pareto tail with tail index 2, meaning $\lim_{\lambda \rightarrow \infty} \lambda^2(1 - G(\lambda))$ is positive and finite.

Proposition 3 showed that a strictly positive fraction of meetings are with traders at any finite upper bound $\bar{\lambda}$. The existence of middlemen is a stronger result, because it implies that this fraction does not vanish as $\bar{\lambda}$ goes to infinity. In order to show the existence of middlemen, we

Proposition 4.

Read: In the limiting economy, the contact-rate distribution has a Pareto tail with tail index 2 and middlemen account for a positive fraction of meetings.

Interpretation: A small measure of extremely high-contact traders can dominate meetings because meetings are weighted by contact intensity.

Economic message: The model provides an analytical foundation for power laws and core-periphery structures in OTC markets.

What not to over-read: Middlemen are a limiting concept; finite markets have very high-contact agents rather than literal infinite-contact agents.

5.5 Proposition 5: Frictionless limit and persistent intermediation

Proposition 5. *Assume $C(\lambda)=c\lambda$. Consider a sequence of limiting equilibria as c converges to zero. The aggregate trading volume $\mathcal{V}$ converges to approximately 2.46γ and can be decomposed as follows: middlemen's purchases from other middlemen account for a volume $\mathcal{V}_{mm}=\frac{1}{2}\gamma$; middlemen's purchases from non-middlemen account for a volume of $\mathcal{V}_{mn}=\frac{1}{2}\gamma$; non-middlemen's purchases from middlemen account for a volume $\mathcal{V}_{nm}=\frac{1}{2}\gamma$; and non-middlemen's purchases from non-middlemen account for a volume $\mathcal{V}_{nn}\approx 0.96\gamma$.*

To prove this proposition, we first compute trading probabilities in an economy with finite $\bar{\lambda}$, then construct a limiting equilibrium, and then take the limit of trade volume as c converges to zero. See the proof of Proposition 5 for details. The proof also provides the exact expression for volume and the fraction of meetings with middlemen.

Proposition 5.

Read: Aggregate trading volume remains large as contact costs vanish, and reallocation runs through middlemen.

Interpretation: Better contact technology does not eliminate intermediaries; it makes fast intermediation chains more effective.

Economic message: Intermediation can remain prominent even when search frictions become small.

What not to over-read: The numerical volume expression is a limiting result, not a direct empirical calibration.

5.6 Planner equations (12)–(15): Social surplus and social value

us the gain the planner enjoys by moving a trader from the misaligned state to the well-aligned state. We prove that this satisfies

$$(r+2\gamma)S(\lambda) = \Delta + \frac{\lambda}{2} \int_{\mathcal{X}} \left(((S(\lambda') - S(\lambda))^+ - (S(\lambda) + S(\lambda'))^+) m_{\lambda'} \right. \\ \left. + ((-S(\lambda) - S(\lambda'))^+ - (S(\lambda) - S(\lambda'))^+) (1 - m_{\lambda'}) \right) d\mu_F(\lambda'). \quad (12)$$

This is the direct counterpart to equation (8). The only difference is that the planner internalizes the full gains from trade in each transaction, whereas an individual trader only internalizes her own half.

Second, we prove that trade occurs if and only if it results in an increase in total social surplus. This gives us the flow balance equation governing the stationary distribution of misalignment:

$$\left(r + \gamma + \frac{\lambda}{2} \int_{\mathcal{X}} (\mathbb{I}_{S(\lambda) + S(\lambda') > 0} m_{\lambda'} + \mathbb{I}_{S(\lambda) > S(\lambda')} (1 - m_{\lambda'})) d\mu_F(\lambda') \right) m_{\lambda} \\ = \left(\gamma + \frac{\lambda}{2} \int_{\mathcal{X}} (\mathbb{I}_{S(\lambda) < S(\lambda')} m_{\lambda'} + \mathbb{I}_{S(\lambda) + S(\lambda') < 0} (1 - m_{\lambda'})) d\mu_F(\lambda') \right) (1 - m_{\lambda}). \quad (13)$$

This is exactly analogous to equation (9), with social surplus playing the role of private surplus.

Third, we define the social value function

$$r\Pi_{\lambda} = -\gamma S(\lambda) \\ + \frac{\lambda}{2} \int_{\mathcal{X}} ((S(\lambda') - S(\lambda))^+ m_{\lambda'} + (-S(\lambda) - S(\lambda'))^+ (1 - m_{\lambda'})) d\mu_F(\lambda') - \bar{\theta}\lambda - rC(\lambda), \quad (14)$$

where

$$\bar{\theta} = \int_{\mathcal{X}} \frac{\gamma - (r+2\gamma)m_{\lambda}}{\lambda} S(\lambda) d\mu_F(\lambda). \quad (15)$$

Planner equations.

Read: The planner values full trading surplus and internalizes meeting externalities.

Interpretation: Comparing planner equations with private equations isolates the wedges behind inefficiency.

Economic message: Middlemen are socially useful, but decentralized contact investment is misallocated.

What not to over-read: The planner is a welfare benchmark rather than an institutional prescription.

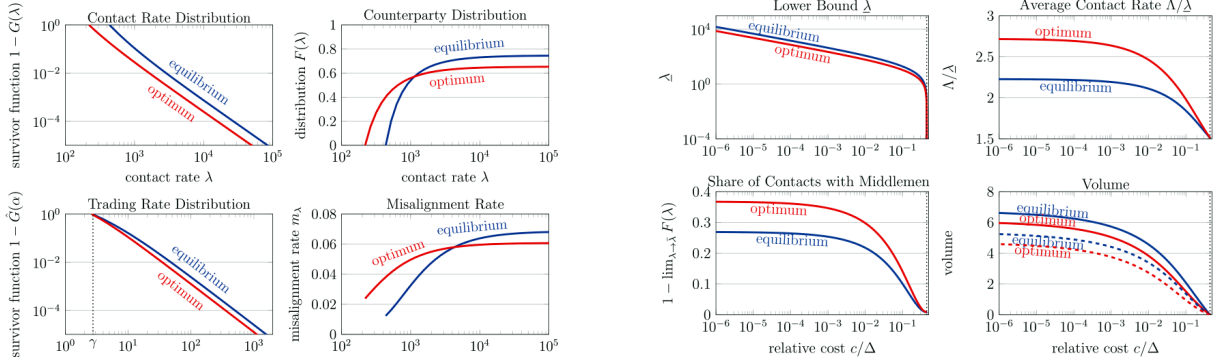
5.7 Figure 2: Equilibrium and optimal counterparty distribution, contact-rate distribution, trading-rate distribution, and misalignment rates

Read: Equilibrium contact rates are too high; both equilibrium and optimum have Pareto tails; fast traders are too misaligned in equilibrium.

Interpretation: The optimum does not flatten the hierarchy. It changes the scale and composition of contact investment.

Economic message: The inefficient margin is not the existence of middlemen but decentralized overinvestment and misallocation of meetings.

What not to over-read: The numerical exercise illustrates forces rather than fitting a specific OTC market.



5.8 Figure 3: Features of the equilibrium and optimal allocation given relative costs

Read: Equilibrium volume exceeds optimal volume over a range of relative contact costs, and the optimum assigns a larger share of contacts to middlemen.

Interpretation: Overinvestment and excessive trading are robust, not artifacts of one chosen cost value.

Economic message: The model gives a rationale for policies that discourage excessive contact investment or trading volume.

What not to over-read: The result does not imply all trade is wasteful; it distinguishes socially useful intermediation from excessive decentralized trading.

6 Short summary

- Ex ante identical traders choose heterogeneous contact rates because intermediation creates dynamic option value.
- Faster traders become endogenous middlemen.
- With linear contact costs, the limiting contact-rate distribution has a Pareto tail with index 2.
- Intermediation persists as contact costs vanish.
- The optimum also uses middlemen, but equilibrium features excessive contact investment and trading.

7 Expanded conclusion

7.1 Core conclusion

Farboodi, Jarosch, and Shimer show that endogenous contact investment can generate the full architecture of OTC market structure: heterogeneous access, intermediation chains, a core of middlemen, and power-law trading rates.

7.2 Economic intuition

A fast trader has a valuable option to absorb a position today because she can sell or buy again quickly tomorrow. This option value makes homogeneous contact rates unstable and produces a continuous hierarchy.

7.3 Incremental contribution

The paper endogenizes what many OTC models assume: dealer status, heterogeneous market access, and core-periphery structure. It also provides a normative benchmark and shows why equilibrium trading can be excessive even though intermediation itself is socially valuable.

7.4 Policy takeaway

The problem is not that middlemen exist. The problem is that decentralized traders do not internalize congestion and thick-market externalities. Policy can improve welfare by discouraging excessive contact investment and trading volume.